

Project Proposal

Project Outline

Project Idea Title: 1. Developing a Mobile App for Local Disaster Preparedness and Response

Module Name: CM3050 Mobile Development

Project Aim: To develop an engaging mobile application that helps with local disaster preparedness through gamification and provides real-time emergency response guidance.

Motivation for the project

Generally, people would assume that Singapore is safe from natural disaster but this type of perception is dangerous. Although Singapore is at low risk from major earthquakes and tsunamis, there are still other disaster that could happen.

There are 2 types of disaster that could happen in Singapore, natural disaster and man-made disaster.

Example of natural disaster:

- Flash flood
- Haze
- Tremors (From earthquake)
- Tsunami (Low risk)
- Pandemic (SARS, Covid-19)

Example of man-made disaster:

- Environmental Pollution (Oil/Gas spills)
- Industrial Accidents (Biochemical spills)
- Urban Fires
- Structural Collapses (Sinkhole/Buildings)
- Terrorist Attack

Comparing apps (Red Cross)

For this type of application, usually it's lacking on engagement which results in low user rates outside actual emergencies. For example, Red Cross Emergency App has comprehensive emergency information but it does not provide any engaging activities like gamification.



Comparing apps (SGSecure and myResponder)

The app that is available in Singapore for emergencies are SGSecure and myResponder but they are fairly limited to man-made disasters. SGSecure is used to report suspicious activity and SOS. It can be used to report emergency such as when someone has fainted. Which is pretty similar to myResponder app which focuses on community responders for medical and fire emergencies.



Personal SOS

If you require emergency Police assistance but are unable to call '999', activate the Personal SOS feature to contact the Police via the Police's SMS 70999 service, and allow them to access your approximate location.



Motivation for the project (cont.)

There hasn't been any mobile application that includes all the necessary functions. Most of the current application in Singapore is not proactive on disaster preparedness.

It would be ideal to develop one. It is important to make it engaging to keep the user interested and focused which will result in a good user retention.

Main functions

- Gamified preparedness (quizzes, badge, achievement, checklist, and weekly challenges)
- Location-based notification system (Integrated with weather API, notifications for nearby disaster, and severity-based alert)
- Resource hub with emergency contacts, response guide, emergency plan builder and supply checklist
- Reporting sightings (could report sudden flood or other emergencies)

Deliverables

- Functional mobile application
- Technical documentation
- Testing report with user feedback
- Project report

Processes, techniques, methodologies, and framework

- Dart programming language
- Flutter framework
- SQLite database
- Flood Alerts API from data.gov.sg
- PSI API from data.gov.sg

Target Audience

- Primary Users would be families which are generally motivated by the safety of family members. Would require simple informations with quick and actionable guidance.
- Secondary Users would be young professionals which are more tech savvy and responsive to gamification. Would generally be motivated community connection or self sufficiency.
- Tertiary Users would be community organizations that could help promote the mobile application and would benefit from coordinated community preparedness efforts.